



Q. 1. Name the physical quantity whose S.I. unit is JC^{-1} . Is it a scalar or a vector quantity?

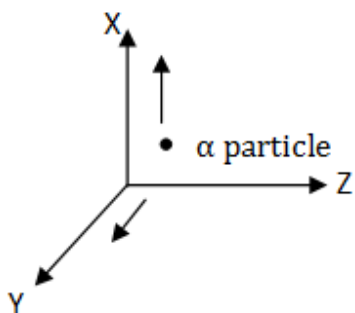
Answer:

We know that work done (W) = Charge (Q) X Potential Difference (V), therefore $V=W/Q=J/C$

Physical Quantity is electrostatic potential (V).

Potential is actually work done, work is scalar so potential is also Scalar quantity

Q. 2. A beam of α – particles projected along +x - axis, experiences a force due to a magnetic field along the +y - axis. What is the direction of the magnetic field?



Answer: We know that force is cross product of magnetic field and velocity

$$\vec{F} = \vec{B} \times \vec{V}$$

Particles projected along X axis i.e velocity, Force is along Y axis therefore Magnetic field must be along Z axis. Since $\vec{F}$ is perpendicular to the plane of $\vec{B}$ and $\vec{V}$, also evident from cross product rule

Q. 3. Define self – inductance of a coil. Write its S.I. units.

Ans. Self inductance:

$$E = -L \Delta I / \Delta t$$

If $\Delta I / \Delta t$ = rate of change of current = 1, then $E=L$ (neglecting negative sign)

Thus Self- inductance of a coil is equal to the e.m.f induced in the coil when rate of change of current through the coil is unity.

Its SI unit is Henry (H).

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Q. 4. A converging lens is kept coaxially in contact with a diverging lens – both the lenses being equal focal lengths. What is the focal length of the combination?

Answer:

Let f_1 is the focal length of convex (converging) lens and

f_2 is the focal length of the concave (diverging) lens.

Equivalent focal length F for the two lenses in contact is given by

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{F} = \frac{1}{f_1} - \frac{1}{f_2} = \frac{f_2 - f_1}{f_1 f_2}$$

[convex lens focal length taken as + ve and concave lens focal length taken as – ve]

$$\therefore F = \frac{f_1 f_2}{f_2 - f_1}$$

Q. 5. Define ionisation energy. What is its value for a hydrogen atom?

Answer:

Definition: The minimum energy required in removing a single electron from the atom or molecule is called Ionisation energy.

Unit of Ionisation energy is Electron Volts (eV).

Depending upon the electron removed we have first ionisation energy, second ionisation energy etc.

1st ionization energy (Atom Ground state) $X \rightarrow X^+ + e^-$

2nd ionization energy (Single ionised state) $X^+ \rightarrow X^{2+} + e^-$

3rd ionization energy (Doubly ionised state) $X^{2+} \rightarrow X^{3+} + e^-$

Ionisation energy for hydrogen atom is 13.6 eV.

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Q. 6. Two conducting wires X and Y of same diameter but different materials are joined in series across a battery. If the number density of electrons in X is twice that in Y, find the ratio of drift velocity of electrons in the two wires.

Answer:

We know that current is related to drift velocity (v_d) by

$$I = n A e v_d$$

v_{dx} = drift velocity in X

v_{dy} = drift velocity in Y

$$n_x = 2n_y$$

Since wires are in series (Current I is same) and X and Y have same diameter (Area A is same)

$$\therefore I_x = I_y$$

$$\therefore n_x A e v_{dx} = n_y A e v_{dy}$$

$$2 n_y A e v_{dx} = n_y A e v_{dy} \Rightarrow \frac{v_{dx}}{v_{dy}} = \frac{1}{2}$$

$$\therefore v_{dx} : v_{dy} = 1 : 2$$

Q. 7. Name the part of electromagnetic spectrum whose wavelength lies in the range of 10^{-10} m. Give its one use.

Answer: We know that wavelength of range 0.01 to 10 nanometre (1 nanometre = 10^{-9} m) falls in X-ray region.

Therefore given wavelength 10^{-10} m = 0.1 nanometre is in X-ray region

Uses: Airport security luggage scanners use X-rays for inspecting the interior of luggage for security threats before loading on aircraft.

Q. 8. When light travels from a rarer to a denser medium, the speed decreases. Does this decrease in speed imply a decrease in the energy carried by the light wave? Justify your answer.

Answer: Frequency of light does not change while passing from rarer to denser medium.

We know that **Energy** $E = h\nu$

h = plank constant

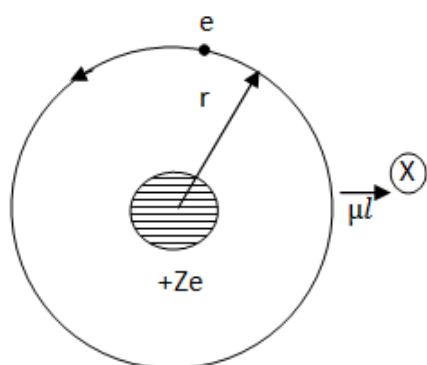
ν = frequency

Since both h and ν remains same hence Energy does not change.



Q. 9. Deduce the expression for the magnetic dipole moment of an electron orbiting around the central nucleus.

Answer:



The electron of charge $(-e)$ executes uniform circular motion around a stationary heavy nucleus of charge $+Ze$. We know that charge in motion constitutes a current. (current = charge/time).

$$\therefore I = \frac{e}{T} \dots \dots (i)$$

Given: T is the time period of revolution.

r be the orbital radius of electron

v the orbital speed.

Therefore distance = speed $\times$ time i.e. $2\pi r = T \times v$

$$T = \frac{2\pi r}{v} \dots \dots (ii)$$

Substituting the value of T in (i), we get

$$I = \frac{ev}{2\pi r}$$

The magnetic moment (*product of current and area* = IA) associated with this circular current is

$$\text{Magnetic Moment} = I \pi r^2 = \frac{ev}{2\pi r} \cdot \pi r^2 = \frac{evr}{2}$$

Magnitude of Magnetic moment = $\frac{evr}{2}$. Now magnetic moment being vector quantity let's find direction.

To find direction magnetic moment: Current is opposite to the motion of electron i.e. if the electron rotates along anticlockwise direction as shown then current is along clockwise direction, rotating the right hand along the direction of current, thumb gives direction perpendicular to the plane of the paper and away from the observer i.e. into the plane of the paper symbolised by cross $\otimes$.

Q. 10. A spherical conducting shell of inner radius r_1 and outer radius r_2 has a charge ' Q '. A charge ' q ' is placed at the centre of the shell.

- What is the surface charge density on the (I) inner surface, (II) outer surface of the shell?
- Write the expression for the electric field at a point $x > r_1$ from the centre of the shell.

Answer:

Surface charge density on inner surface = $\frac{q}{4\pi r_1^2}$



Surface charge density on the outer surface $= \frac{q+Q}{4\pi r_2^2}$

Let the point P lie outside the charged shell $OP = x > r_1$

We know that flux through the surface

$$\phi = \oint_s d\phi = \oint_s E dA = E \cdot 4\pi x^2 \rightarrow (1)$$

Applying the Gauss's theorem the flux through the Gaussian surface

$$\phi = \frac{\text{Charged enclosed}}{\epsilon} = \frac{Q+q}{\epsilon} \rightarrow (2)$$

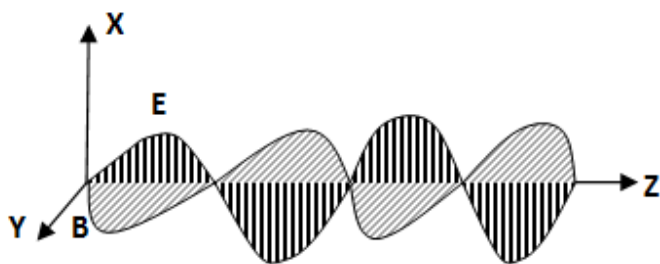
Equating (1) and (2):

$$E \cdot 4\pi x^2 = \frac{Q+q}{\epsilon}$$

$$E = \frac{1}{4\pi\epsilon} \frac{Q+q}{x^2} \rightarrow (3)$$

Q. 11. Draw a sketch of a plane electromagnetic wave propagating along the z – direction. Depict clearly the directions of electric and magnetic fields varying sinusoidally with z.

Answer: We know that direction of propagation is perpendicular to the plane of drawn through Electric and Magnetic Vector. Hence if Electric vector drawn along X axis, Magnetic vector along Y axis then propagation will be perpendicular to X-Y plane i.e. along Z axis as shown.





Q. 12. Show that the electric field at the surface of a charged conductor is given by $\vec{E} = \frac{\sigma}{\epsilon_0} \hat{n}$, where σ is the surface charge density and $\hat{n}$ is a unit vector normal to the surface in the outward direction.

Answer:

Given : σ = surface density of charge i.e. charge per unit surface area

ϵ = permittivity of the surrounding medium

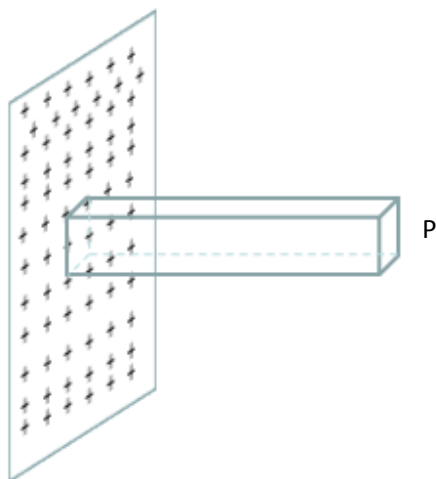
P is the given point close to the surface of the charged plate.

Let E be the intensity at P ?

Let us imagine a rectangular box passing through the points P. This closed surface is our Gaussian surface and is known as pill box.

Let A = Area of the side face of pillbox. Then Enclosed Charge = σA . Hence by Gauss's theorem

$$\phi = \frac{\sigma A}{\epsilon} \rightarrow (1)$$



Since electric lines of force do not exist inside charged plate hence we have only one side of pill box i.e. outward. The lines of force are perpendicular to the end faces only.

$$\phi = 0 + 0 + 0 + 0 + EA = EA \rightarrow (2)$$

From (1) and (2)

$$EA = \frac{\sigma A}{\epsilon}$$

$$E = \frac{\sigma}{\epsilon}$$

Direction is away from the plate and perpendicular to the plane of side face of pill box as shown in the figure.

$$\vec{E} = \frac{\sigma}{\epsilon_0} \hat{n}$$

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Q. 13. Two identical loops, one of copper and the other of aluminium, are rotated with the same angular speed in the same magnetic field. Compare (I) the induced emf and (II) the current produced in the two coils. Justify your answer.

Answer:

We know that induced emf (e) and induced current (i) in a rotating coil is given by

$$e = NAB\omega \sin \omega t, \quad e = e_p \sin \omega t, \quad \text{Hence peak value of emf} = e_p = NAB\omega.$$

$$\text{Similarly peak value of current} = i_p = e_p / R$$

Where

N = number of turns

A = Area of the coil

B = Magnetic Field

ω = Angular velocity

Since all are same for both copper and aluminium, and does not depend on nature of material hence

(i) The ratio of induced emf = 1 : 1

(ii) The ratio of induced current = 1 : 1

Q. 14. An α – particle and a proton are accelerated from rest by the same potential. Find the ratio of their de Broglie wavelengths.

Answer: We know that de Broglie wavelength (λ) is given by $\lambda = \frac{h}{mv}$

We know that charge $\times$ potential = kinetic energy therefore $eV = \frac{1}{2}mv^2$

$$\text{or } 2meV = m^2v^2 \text{ or } \sqrt{2meV} = mv$$

$$\text{Therefore } \lambda = \frac{h}{\sqrt{2meV}} \therefore \lambda \propto \frac{1}{\sqrt{m}} \quad [\text{For same potential} = V = \text{constant}]$$

We know that alpha particle = ${}^4_2\text{He}$

Proton is ${}^1_1\text{H}$

$\therefore$ Mass of alpha particle = 4 times mass of proton

$$\therefore \lambda_{\text{proton}} \propto \frac{1}{\sqrt{m}} \text{ and } \lambda_{\text{alpha}} \propto \frac{1}{\sqrt{4m}}$$

$$\therefore \frac{\lambda_{\text{proton}}}{\lambda_{\text{alpha}}} = \frac{1}{\sqrt{m}} \times \frac{\sqrt{4m}}{1} = \frac{2}{1}$$

$$\therefore \lambda_{\text{proton}} : \lambda_{\text{alpha}} = 2 : 1$$

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Q. 15. Write two factors justifying the need of modulating a signal.

A carrier wave of peak voltage 12 V is used to transmit a message signal. What should be the peak voltage of the modulating signal in order to have a modulation index of 75%?

Answer:

1. To decrease the antenna size

Explanation: We know that for transmitting a signal of wavelength λ the antenna height required is $\lambda/4$.
Therefore to transmit $\nu = 1$ Hz frequency

$$c = \lambda \times \nu$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$3 \times 10^8 = \lambda \times 1$$

$$\lambda = 3 \times 10^8 \text{ m}$$

Height of antenna required $(h) = \lambda/4 = 3 \times 10^8 / 4 = 300 \times 1000 \times 1000 / 4 = 75000,000 \text{ m} = 75000 \text{ KM}$, Thus it is difficult to construct such huge antenna. Now let us modulate this signal to high frequency say 100 MHz then

$$3 \times 10^8 = \lambda \times 100 \times 10^6$$

$$\lambda = 3 \text{ m}$$

Therefore required antenna height $= \lambda/4 = 3/4 = 0.75 \text{ m}$, which is easy to construct.

2. To simultaneously send data and to multiplex data.

Explanation: without multiplexing Signal, it will interfere with the normal voice in the air

Problem: We know that modulation index $\mu_a = \frac{A_m}{A_c}$

$$\text{Given } \mu_a = 72\% = 0.75$$

$$\mu_a = 72\% = 0.75$$

$$0.75 = \frac{A_m}{12}$$

$$\therefore A_m = 0.75 \times 12 = 9.00 = 9 \text{ V}$$

Q. 16. Write Einstein's photoelectric equation. State clearly the three salient features observe in photoelectric effect, which can be explained on the basis of the above equation.

Answer: Einstein's photoelectric equation: Supplied energy to photo electron is used up to come out (work function ϕ_0), remaining is used as kinetic energy.

$$\text{i.e. } h\nu = \phi_0 + \frac{1}{2} m v_{\text{max}}^2$$

$$K_{\text{max}} = \frac{1}{2} m v_{\text{max}}^2 = h\nu - \phi_0 = h\nu - h\nu_0$$

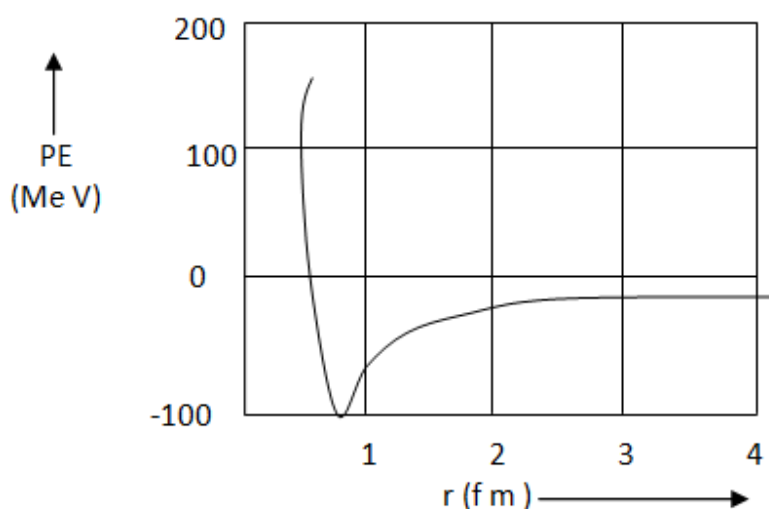


- I. A part of the energy of the photon is used in liberating the electron from the metal surface which is equal to the work function ϕ_0 of the metal.
- II. The rest of the energy of the photon is used to get the maximum kinetic energy $K_{\max}$ to the emitted photo electron.
- III. The negative potential on the collector at which the photoelectric current becomes zero is called the stopping potential. Stopping voltage varies linearly with frequency of light, but depends on the type of material. $K_{\max} = h(\nu - \nu_0)$

Q. 17. Draw a plot of potential energy of a pair of nucleons as a function of their separation. Write two important conclusions which you can draw regarding the nature of nuclear forces.

Answer:

Along Y-axis Plot of potential energy in MeV is taken and along X-axis distance of separation for the pair of nucleons in femtometer(fm) taken



Nature of Nuclear Force:

- I. The nuclear force between two nucleons falls rapidly to zero as their distance is more than a few femtometres (approx. = 0.8 femtometers)
- II. For a separation greater than $(0.8 \text{ fm approx}) r_0$, the force is attractive and for separations less than r_0 , the force is strongly repulsive.

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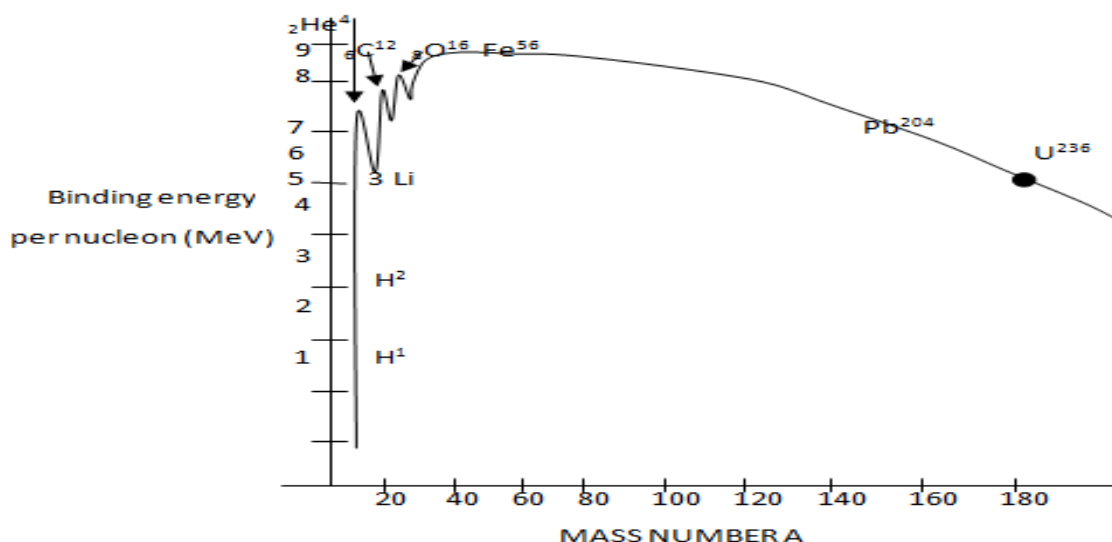


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Or,

Draw a plot of the binding energy per nucleon as a function of mass number for a large number of nuclei, $2 \leq A \leq 240$. How do you explain the constancy of binding energy per nucleon in the range $30 < A < 170$ using the property that nuclear force is short – ranged?

Answer: Binding energy per nucleon is taken along Y-axis and mass number A is taken along X-axis, nuclei having mass number between 2 and 240 ($2 \leq A \leq 240$) considered, the required graphical representation is as shown.



The nucleus in the consistent Binding energy region is large enough so that nuclear forces no longer completely extend efficiently across its width. Attractive nuclear forces in this region, as atomic mass increases, are nearly balanced by repellent electromagnetic forces between protons, as the atomic number increases.

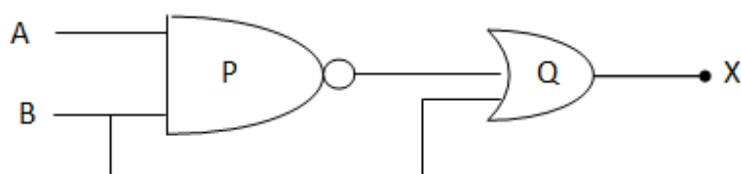
The constancy of the binding energy in the range $30 < A < 170$ is due to fact that the nuclear force is short – ranged.

In elements heavier than xenon, there is a decrease in binding energy per nucleon as atomic number increases. In this region of nuclear size, electromagnetic repulsive forces are beginning to overcome the strong nuclear force attraction.



Q. 18.

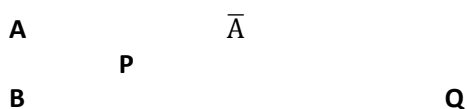
- I. Identify the logic gates marked P and Q in the given logic circuit.



- II. Write down the output at X for the inputs A = 0, B = 0 and A = 1, B = 1.

Answer:

- I. The logic gate symbol denoted by P is NAND gate and the logic gate symbol denoted by Q is OR gate.



Finally output from Q gate is $X = \bar{A} + B$.

- II. Truth Table :-

Input		$\bar{A}$	Output
A	B		$X = \bar{A} + B$
0	0	1	1
1	1	0	1

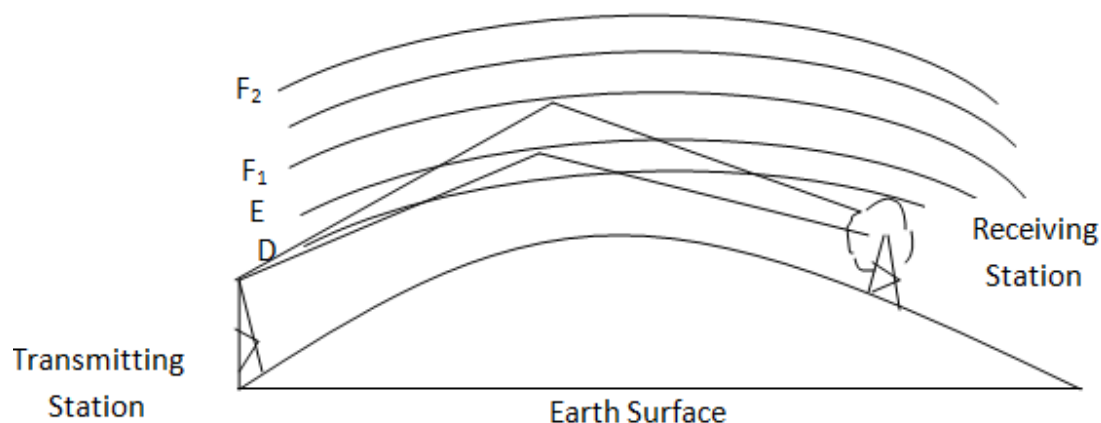
Finally output from Q is always 1 (TRUTH) as indicated from the above Truth Table.



Q. 19. Which mode of propagation is used by short wave broadcast services having frequency range from a few MHz up to 30 MHz? Explain diagrammatically how long distance communication can be achieved by this mode. Why is there an upper limit to frequency of waves used in this mode?

Answer: For short wave broadcast services having frequency range from a few MHz up to 30 to 40 MHz , Sky wave propagation is used.

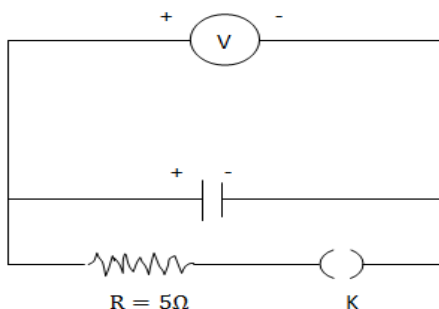
Diagrammatic explanation: Long distance communication achieved through ionosphere reflection of radio waves, which after reflection from layers sends the waves back to the earth as shown in the figure.



Ionosphere layer extends up to 400 miles above earth surface (various sub layers are D layer: 30-60 miles, E layer: 60-70 miles, F_1 layer: 100-140 miles, F_2 layer: 200-300 miles). This mode of propagation is called sky wave propagation. Electromagnetic wave of frequencies higher than 30 MHz penetrate the ionosphere and escape.



Q. 20. Write any two factors on which internal resistance of a cell depends. The reading on a high resistance voltmeter, when a cell is connected across it, it reads 2.2 V. When the terminals of the cell are also connected to a resistance of 5Ω as shown in the circuit, the voltmeter reading drops to 1.8 V. Find the internal resistance of the cell.



Answer: Factors affecting internal resistance of a cell are

1. The nature of the electrolyte.
2. The nature of electrodes.
3. Distance between the electrodes.
4. Area of the electrodes, immersed in the electrolyte. If area increases, internal resistance decreases.

Problem: Let r = internal resistance of the cell.

$$I = \frac{E}{r} \therefore I = \frac{2.2}{r} \dots \dots \dots (i)$$

When the terminals of the cell are connected to a resistance of 5Ω then

$$I = \frac{1.8}{5+r}$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} \therefore R = \frac{R_1 R_2}{R_1 + R_2}$$

$$\therefore I = 1.8 \times \left(\frac{5+r}{5r} \right) \dots \dots \dots (ii)$$

Solving equation (i) and (ii), we get

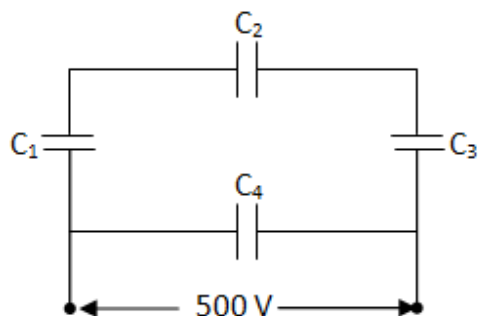
$$\frac{2.2}{r} = 1.8 \times \left(\frac{5+r}{5r} \right) \Rightarrow \frac{2.2}{1.8} = \frac{5+r}{5}$$

$$\Rightarrow \frac{11}{9} = \frac{5+r}{5} \Rightarrow 45 + 9r = 55$$

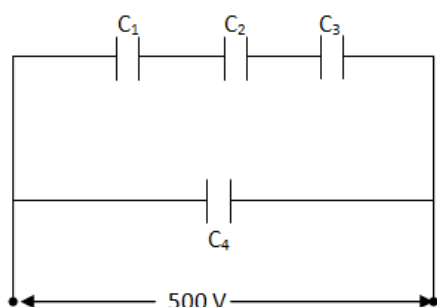
$$\Rightarrow 9r = 55 - 45 \therefore r = \frac{10}{9} \Omega$$



Q.21. A network of four capacitors each of $12\mu\text{F}$ capacitance is connected to a 500 V supply as shown in the figure. Determine (a) equivalent capacitance of the network and (b) charge on each capacitor.



Answer: From the figure it is clear that C_1, C_2, C_3 are in series and this combination is in parallel to C_4 therefore above circuit can be simplified with the equivalent network of the circuit as



The capacitors C_1, C_2 and C_3 are in series

$$\therefore \frac{1}{C_{\text{Series}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \Rightarrow \frac{1}{C_{\text{Series}}} = \frac{1}{12} + \frac{1}{12} + \frac{1}{12}$$

$$\frac{1}{C_{\text{Series}}} = \frac{3}{12} \Rightarrow C_{\text{Series}} = 4\ \mu\text{F}$$

Now C_{Series} and C_4 are in parallel combination.

(a) Therefore the equivalent capacitance of the network is $C = C_{\text{Series}} + C_4 = 4 + 12 = 16\ \mu\text{F} = 16 \times 10^{-6}\text{ F}$

(b) Total charge of capacitor $q = CV$, $q = 16 \times 10^{-6} \times 500 = 8000 \times 10^{-6} = 8000\ \mu\text{C}$

Charge on capacitor C_4 is $q = C_4 V = 12 \times 10^{-6} \times 500 = 6000 \times 10^{-6}\text{ C} = 6000\ \mu\text{C}$

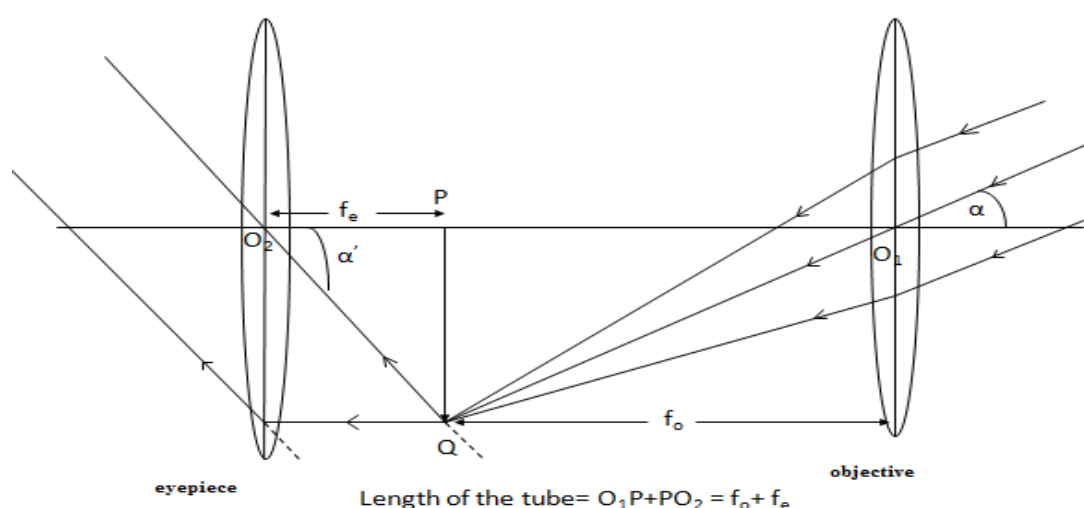
Charge on each capacitor C_1, C_2 and C_3 is $q = C_s V = 4 \times 10^{-6} \times 500 = 2000 \times 10^{-6}\text{ C} = 2000\ \mu\text{C}$



Q. 22.

- I. Draw a neat labelled ray diagram of an astronomical telescope in normal adjustment. Explain briefly it's working.
- II. An astronomical telescope uses two lenses of powers 10 D and 1 D. What is its magnifying power in normal adjustment?

Answer: Ray diagram of an astronomical telescope (normal adjustment):



Working: Astronomical Telescope has two lenses an objective (focal length f_o) and eyepiece (focal length f_e). The objective has a large focal length and much larger aperture than the eyepiece ($f_o > f_e$). When parallel rays from astronomical bodies at infinity incident on the objective after refraction at objective a real image is formed at the focal plane of objective at a distance f_o . For the eyepiece this image formed by objective serves as object, in normal adjustment the distance of the eyepiece is such that this PQ object is at its focal plane hence finally eyepiece forms the final inverted image at infinity as shown.

II. We know that $P = 1/f$, the power of lens given = 10 D $\therefore$ focal length = $\frac{100}{10} = 10$ cm.

The power of lens = 1 D, focal length = $\frac{100}{1} = 100$ cm.

Power in diopetre is converted to cm hence multiplied by 100.

We know that in astronomical telescope $f_o > f_e$. Therefore $f_o = 100$ cm and $f_e = 10$ cm

Magnifying power = $\frac{f_o}{f_e} = \frac{100}{10} = 10$



Q. 23. In Young's double slit experiment, the two slits 0.15 mm apart are illuminated by monochromatic light of wavelength 450 nm. The screen is 0.1 m away from the slits.

- Find the distance of the second (I) bright fringe, (II) dark fringe from the central maximum.
- How will the fringe pattern change if the screen is moved away from the slits?

Answer: Given distance between two slits(d) = 0.15 mm = $0.15 \times 10^{-3} \text{ m} = 15 \times 10^{-5} \text{ m}$

$$\lambda = 450 \text{ nm} = 450 \times 10^{-9} \text{ m} = 4.5 \times 10^{-7} \text{ m},$$

$$\text{Distance of the screen from the light source}(D) = 1.0 \text{ m}$$

- (i) We know that for n th bright fringe

$$x_n = \frac{n\lambda D}{d}, \text{ distance of the second bright fringe}(n=2) = x_2 = \frac{2\lambda D}{d}$$

$$= \frac{2 \times 4.5 \times 10^{-7} \times 1.0}{15 \times 10^{-5}} = \frac{2 \times 4.5}{15} \times 10^{-2} = 0.6 \times 10^{-2} \text{ m} = 6 \text{ mm}$$

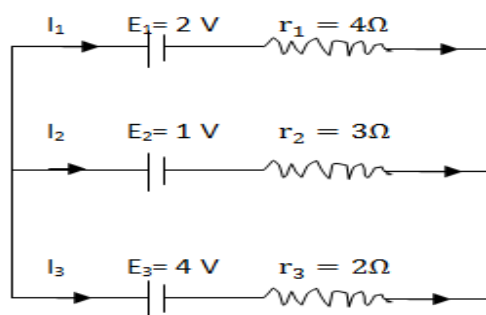
$$(ii) \text{ Distance of the second dark fringe} = x_n = \frac{(2n-1)\lambda D}{2d}, \quad x_2 = \frac{3\lambda D}{2d}$$

$$= \frac{3 \times 4.5 \times 10^{-7} \times 1.0}{2 \times 15 \times 10^{-5}} = \frac{3 \times 4.5}{30} \times 10^{-2} = 4.5 \times 10^{-2} \text{ m} = 4.5 \text{ mm}$$

- We know that fringe width (β) = $\frac{\lambda D}{d}$

When screen is moved away, D increases. Therefore width of the fringes increases but the angular separation ($\frac{\lambda D}{d}$) remains the same.

Q. 24. State Kirchhoff's rules. Use these rules to write the expressions for the currents I_1 , I_2 and I_3 in the circuit diagram shown.





Answer: In complicated networks to find current flowing through different branches we apply Kirchhoff's law.

First law: The algebraic sum of currents meeting at a point is zero. i.e. $\sum i = 0$.

In the given circuit, Considering outward direction of current as positive and inward as negative then at the point F

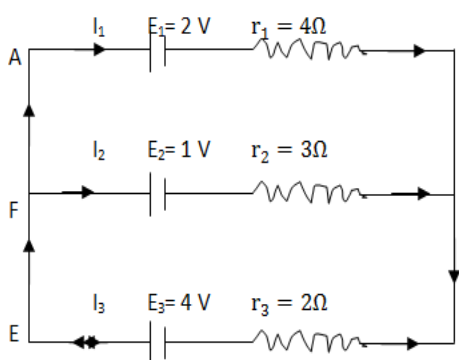
According to Kirchhoff's first law, at junction F

$$I_1 + I_2 = I_3$$

Second law: In a closed mesh of electrical conductors the algebraic sum of product of resistance and the respective current in the different branches is equal to the total e.m.f applied in the closed mesh.

$$\sum iR = \sum E$$

The given circuit diagram is



According to Kirchhoff's second law, in closed circuit ABCFA, we have

$$-2 - 4I_1 + 3I_2 + 1 = 0 \quad [\because \sum IR + \sum E = 0]$$

$$-4I_1 + 3I_2 - 1 = 0 \quad \dots \dots \dots (1)$$

C According to Kirchhoff's second law, in closed circuit FCDEF, we have

$$-1 - 3I_2 - 2I_3 + 4 = 0$$

$$D \quad -3I_2 - 2I_3 + 3 = 0 \quad \dots \dots \dots (2)$$

According to Kirchhoff's first law, at junction F,

$$I_1 + I_2 = I_3 \quad \dots \dots \dots (3)$$

On solving eqns. (1), (2) and (3), we get

$$I_1 = \frac{2}{13}, I_2 = \frac{7}{13} \text{ and } I_3 = \frac{9}{13}$$



Q. 25. a. Write symbolically the β^- decay process of $^{32}_{15}\text{P}$.

b. Derive an expression for the average life of a radionuclide. Give its relationship with the half life.

Answer: (a) $^{32}_{15}\text{P} \rightarrow ^{32}_{16}\text{S} + \beta^-$

We know that β^- is equivalent electron (-1 atomic number reduction but changes in mass number) $^{32}_{15}\text{P} \rightarrow ^{32}_{16}\text{S} + ^0_{-1}\text{e}$

(b) **Average or Mean Life:** Average life is defined as the total life time of all the atoms of the element divided by the total number of atoms present in the sample of the element.

By definition, average life time of the radioactive element is $\tau = \frac{\int_0^{N_0} t \, dN}{N_0}$

Let N_0 = The number of radio active atoms present in the sample at an instant of time $t = 0$

N = The number of radio active atoms present in the sample at an instant of time t

$N - dN = N_0$ = The number of radio active atoms present in the sample at an instant of time $t + dt$

dN = The number of radio active atoms disintegrating in time dt at an instant of time t

$$\therefore -\frac{dN}{dt} \propto N$$

$$\frac{dN}{dt} = -\lambda dt \rightarrow (1)$$

λ = constant of proportionality known as decay constant or disintegration constant.

$$\frac{dN}{N} = -\lambda dt$$

Integrating both sides :

$$\log_e N = -\lambda t + A \rightarrow (2)$$

Where A is constant of integration. At $t = 0$, $N = N_0$ from equation (2)

$$\log_e N_0 = -\lambda \times 0 + A$$

$$\text{or } A = \log_e N_0 \rightarrow (3)$$

Putting equation (3) in (2)

$$\log_e N = -\lambda t + \log_e N_0$$

$$\log_e \frac{N}{N_0} = -\lambda t$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$N = N_0 e^{-\lambda t} \rightarrow (4)$$

Using equation (4) the number of radio active atoms present at any instant t can be calculated.



$$N = N_0 e^{-\lambda t}$$

From equation we find that at $t = 0$, $N = N_0$ and $t = \infty$, $N = 0$

Half life period of a radioactive element is defined as the time during which half of the radioactive atoms present in the specimen disintegrated.

Let at $t = 0$, $N = N_0$ be the number of radio active atoms present in the sample.

at $t = T$, $N = \frac{N_0}{2}$ = Half of the original radio active atom present in the sample, then T is known as half life.

From equation (1) : $\frac{N_0}{2} = N_0 e^{-\lambda T}$

Taking log of both sides $-\log_e 2 = -\lambda T$

$$T = \frac{\log_e 2}{\lambda} = \frac{0.693}{\lambda}$$

Equation gives the half life period.

When $N = 0$ then $t = \infty$ and when $N = N_0$ then $t = 0$

$$\therefore \tau = \frac{\int_0^\infty t \cdot -\lambda N_0 e^{-\lambda t} dt}{N_0} = \lambda \int_0^\infty t e^{-\lambda t} dt$$

$$\tau = \lambda \left[\frac{t e^{-\lambda t}}{\lambda} - \frac{e^{-\lambda t}}{\lambda^2} \right]_0^\infty \quad (\text{Using integration by parts})$$

$$\tau = \lambda \left[(0 - 0) - \left(0 - \frac{1}{\lambda^2} \right) \right] = \lambda \times \frac{1}{\lambda^2} = \frac{1}{\lambda}$$

Since $\lambda = \frac{0.693}{T}$

$$\therefore \tau = \frac{1}{\frac{0.693}{T}} = \frac{T}{0.693} = 1.44 T$$

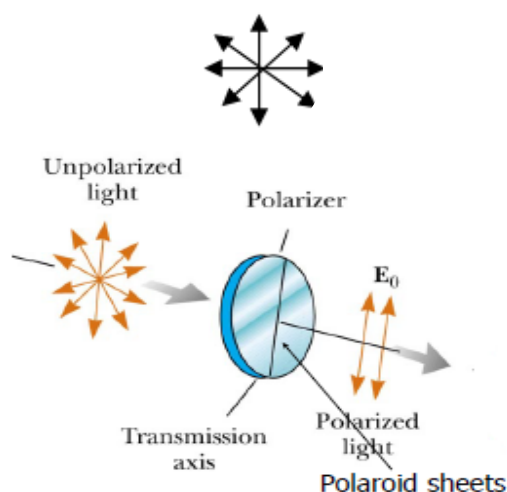
Thus the average life period of a radioactive element (τ) is 1.44 times the half life period (T) of the element.



Q. 26. How does an unpolarized light get polarized when passed through a polaroid?

Two polaroids are set in crossed positions. A third polaroid is placed between the two making an angle θ with the pass axis of the first polaroid. Write the expression for the intensity of light transmitted from the second polaroid. In what orientations will the transmitted intensity be (I) minimum and (II) maximum?

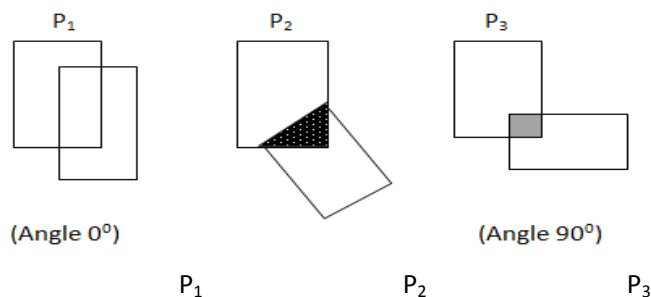
Answer: In unpolarized light, E-field has random directions of vibration.



Linearly polarized light: E-field has one spatial orientation.

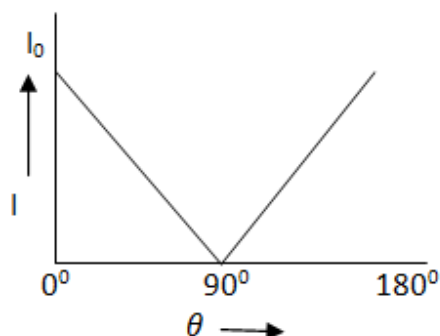
Unpolarized light after passing through polaroid becomes polarized light. When unpolarized light wave pass through Polaroid the light get linearly polarized with the electric vector oscillating along a direction perpendicular to the aligned molecules.

If the light from an ordinary source (like a sodium lamp) passes through a polaroid sheet P_1 , it is observed that its intensity is reduced by half. Rotating P_1 has no effect on the transmitted beam and transmitted intensity remains constant. Considering polaroid P_2 placed before P_1 , though light from the lamp is reduced in intensity on passing through P_2 however on rotating P_1 there is no effect on the light coming from P_2 . In one position, the intensity transmitted by P_2 followed by P_1 is nearly zero. When turned by 90° from this position, P_1 transmits nearly the full intensity emerging from P_2 as shown in the figure.





Graph : The graph between intensity of light and the angle between polariser and analyser is shown as :



When the polaroid is rotated in the path of plane polaroid light, its intensity will vary from maximum (When the vibrations of the plane polarized light are parallel to the axis of the polaroid) to minimum (When the direction of vibrations become perpendicular to the axis of the crystal).

Let I_0 be the intensity of polarized light after passing through first polariser P_1 . Then intensity of light after passing through second polariser P_2 is given by $I = I_0 \cos^2 \theta$.

Q. 27. An illuminated object and a screen are placed 90 cm apart. Determine the focal length and nature of the lens required to produce a clear image on the screen, twice the size of the object.

Answer:

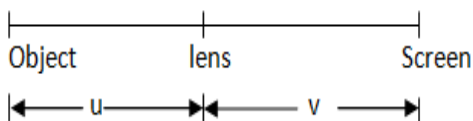
$$\text{Given: } \therefore u + v = 90 \dots \dots \dots (I)$$

$$m = \frac{v}{u} \Rightarrow 2 = \frac{v}{u} \Rightarrow v = 2u \dots \dots (II)$$

Putting the value of v in (I), we get

$$u + 2u = 90 \Rightarrow u = \frac{90}{3} = 30$$

$$\therefore v = 2 \times 30 = 60$$



Using Lens formula

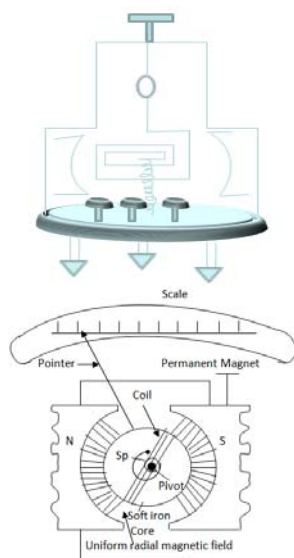
$$\begin{aligned} \frac{1}{f} &= \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{f} = \frac{1}{60} - \frac{1}{-30} \\ &\Rightarrow \frac{1}{f} = \frac{1}{60} + \frac{1}{30} = \frac{3}{60} \\ &\Rightarrow f = \frac{60}{3} = 20 \text{ cm.} \end{aligned}$$

Q. 28.



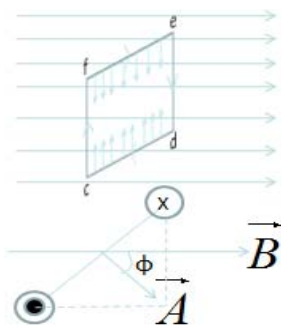
- With the help of a diagram, explain the principle and working of a moving coil galvanometer.
- What is the importance of a radial magnetic field and how is it produced?
- Why is it that while using a moving coil galvanometer as a voltmeter a high resistance in series is required whereas in an ammeter a shunt is used?

Answer : (a) **Moving coil galvanometer:** It is a device used for the detection and measurement of small electric current.



Principle : (Based on the principle of the force of interaction between the current carrying conductor and a magnetic field) A coil is suspended in a magnetic field. The current to be measured is passed through the coil. The current through the coil produces a magnetic field which interacts with the given magnetic field in which the coil is suspended. Due to the interaction the coil gets deflected and measuring the angle of rotation current can be calculated.

A circular or rectangular coil of about 10 to 15 turns of a fine insulated copper or aluminium wire is suspended from a torsion head T, by means of quartz fibre in between the concave pole pieces of a strong magnet. The lower end of the coil is attached to a light springs which brings the coil back to its original position when the current is stopped. The suspension wire and the spring are connected with two terminal screws at the base which acts as the leads of current. The whole thing is enclosed in a metal box provided with glass face and is supported on levelling screws. The angle of rotation of the coil is measured by lamp & scale arrangement, reflection takes place from the tiny mirror M attached with the suspension wire.



Theory : Given n = the number of turns in the coil

l & b = the length of each vertical & horizontal side of the coil respectively (assuming) the coil to be rectangular
 i = the current flowing through the coil.

B = the induction vector of the magnetic field in which the coil is suspended.

We know that the force experienced by the current carrying conductor placed in a magnetic field is

$$\vec{F} = i(\vec{l} \times \vec{B}) \rightarrow (1)$$



Force on each wire 'de' and 'fc'

$$|\vec{F}| = ibB \sin(90 - \phi) = ibB \sin \phi$$

where ϕ = angle between the normal to the plane of the coil and the direction of the magnetic field.

The direction of the force on the wires 'de' & 'fc' are in the plane of the coil along the downward and upward direction respectively and hence cancel out.

The force on the two vertical wires 'cd' and 'ef' by using equation(1) are found to have magnitude

$$F = iB \sin 90^\circ = ilb$$

Applying the right hand curl rule for vector product the directions of the force are as shown in the figure. These two forces being equal in magnitude, opposite in direction parallel and non co-planer constitute a couple.

Moment of the couple $\Gamma = iB \times OC$

$$\Gamma = iBb \sin \phi = i(lb)B \sin \phi = iAB \sin \phi$$

Since there are n turns in the coil the total torque experienced by the current carrying coil in the magnetic field

$$\Gamma = niAB \sin \phi$$

Due to this torque the coil rotates a restoring couple due to the torsion rigidity sets in and when the restoring couple equals to the deflecting couple the coil comes to equilibrium



Let c = torsional couple per unit twist of the suspension wire

θ = the angle through which the coil rotate in the position of equilibrium

Restoring couple = $c\theta \rightarrow (1)$

From equation(2) and (3): $nAiB\sin\phi = c\theta$

$$i = \frac{c}{nAB} \frac{\theta}{\sin\phi} \rightarrow (4)$$

By using concave pole pieces the magnetic field is made

radial so that for any position of coil ϕ is always 90° ,

$$\sin\phi = \sin 90^\circ = 1$$

$$\therefore i = \frac{c}{nAB} \theta \rightarrow (5)$$

Since c, n, A & B are all constant

$$i \propto \theta$$

- b) **Radial magnetic field:** The plane of the coil always lies in the direction of the magnetic field. A radial magnetic field is produced by (A) properly cutting the magnetic pole pieces in the shape of

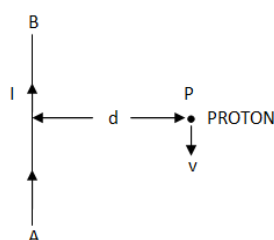


concave faces and (B) using a soft iron core within the coil (shown in the above figure)

- c) A galvanometer can be converted into a voltmeter by connecting a high resistance in series with galvanometer to draw a very small current.
A galvanometer can be converted into an ammeter by connecting a low resistance shunt in parallel with galvanometer to draw large value of current. This low resistance is called Shunt.

Or,

- Derive an expression for the force between two long parallel current carrying conductors.
- Use this expression to define SI unit of current.
- A long straight wire AB carries a current I . A proton P travels with a speed v_1 parallel to the wire, at a distance d from it in a direction opposite to the current as shown in the figure. What is the force experienced by the proton and what is its direction?

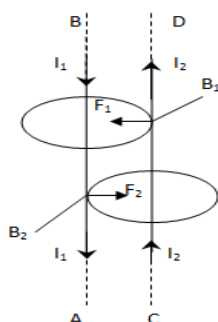


Answer: (a) Force between two parallel current carrying wires

Given: AB and CD are two infinitely long conductors, placed parallel to each other

r = distance of separation between two wires.

I_1 and I_2 = current flowing through AB and CD respectively (direction of current taken same).



Magnetic field produced by current I_1 at any point on CD is

$$B_1 = \frac{\mu_0 I_1}{2\pi r}$$

This field acts perpendicular to CD and into the plane of paper. It exerts a force on wire CD carrying current I_2 is

$$F = B_1 I_2 l$$

$$F = \frac{\mu_0}{2\pi r} \times I_1 \cdot I_2 l$$

$$F = \frac{\mu_0}{2\pi} \cdot \frac{I_1 I_2}{r} \cdot l$$

$\therefore$ Force per unit length is

$$F = \frac{\mu_0}{2\pi} \cdot \frac{I_1 I_2}{r}$$

$$\therefore F = \frac{\mu_0}{4\pi} \cdot \frac{2I_1 I_2}{r}$$

By Fleming's left hand rule, this force will be attractive.

(b) Definition of ampere:

Let $I_1 = I_2 = 1$ A, $r = 1$ m then

$$F = \frac{4\pi \times 10^{-7} \times 2 \times 1 \times 1}{4\pi \times 1} [\because \mu_0 = 4\pi \times 10^{-7}]$$

$$F = 2 \times 10^{-7} \text{ Nm}^{-1}$$

Thus, one ampere can be defined as the amount of current which when flows through each of the two parallel uniform long linear conductors placed in free space at a distance of one metre from each other will attract or repel each other with a force of 2×10^{-7} N per metre of their length.



(c)Magnetic field induction at the point P is

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2I}{r}$$

The direction of $\vec{B}$ is perpendicular to the plane of paper directed inwards
(According to Right Hand Thumb Rule).

The force on moving proton of charge q due to magnetic field B is

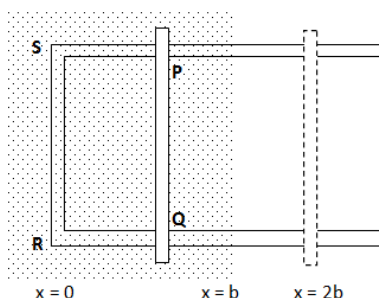
$$F = q v B \sin 90^\circ = q v B \quad (\text{Since } \vec{F} = q(\vec{V} \times \vec{B}))$$

Given : Charge of proton $q = 1.6 \times 10^{-9} \text{ C}$

$$\therefore F = 1.6 \times 10^{-9} \times v B \text{ N}$$

The direction of force on proton, according to Fleming's Left Hand Rule acts in the plane of paper towards right.

Q. 29. State Faraday's law of electromagnetic induction. Figure shows a rectangular conductor PQRS in which the conductor PQ is free to move in a uniform magnetic field B perpendicular to the plane of the paper. The field extends from $x = 0$ to $x = b$ and is zero for $x > b$. Assume that only the arm PQ possesses resistance r . When the arm PQ is pulled outward from $x = 0$ with constant speed v , obtain the expressions for the flux and the induced emf. Sketch the variations of these quantities with distance $0 \leq x \leq 2b$.



Answer: Faraday's law of induction:

First law: The induced emf is proportional to the number of turns in the coil.

$e \propto N \rightarrow (1)$ where N = number of turns in the coil.

Second law: The emf induced across the coil is proportional to the rate of change of flux through the coil.

$e \propto \frac{d\phi}{dt}$ where $\frac{d\phi}{dt}$ is rate of change of flux at an instant t

Combining first and second law : $e \propto N \frac{d\phi}{dt}$

$$e = KN \frac{d\phi}{dt}$$

Where K = constant of proportionality, by proper choice of unit

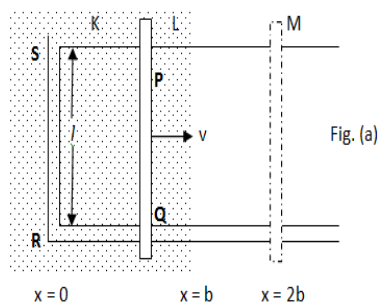
i.e. if e is in volt and flux in tesla - m^2 then constant $K = 1$

$$e = N \frac{d\phi}{dt}$$

Equation can be more correctly written as

$$e = -N \frac{d\phi}{dt}$$

The negative sign indicates that the direction of the induced emf is opposite to that of the cause.



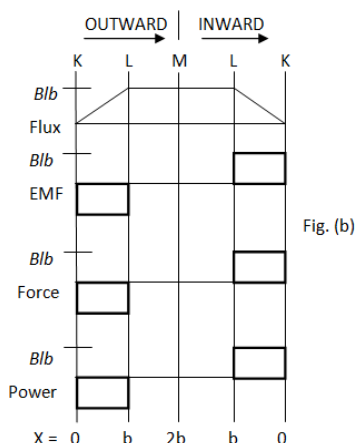
Considering the forward motion from $x = 0$ to $x = 2b$. The flux ϕ_B linked with the circuit SPQR is

$$\begin{aligned}\phi_B &= Blx & 0 \leq x < b \\ &= Blb & b \leq x < 2b\end{aligned}$$

The induced emf is, $\varepsilon = -\frac{d\phi_B}{dt}$ (as number of turns $N=1$)

$$\begin{aligned}\varepsilon &= -Blv & 0 \leq x < b \\ &= 0 & b \leq x < 2b\end{aligned}$$

When the induced emf is non-zero, the current I is (in magnitude) $I = \frac{Blv}{r}$



The force required to keep the arm PQ in constant motion is $l^2 B$. Its direction is to the left. In magnitude

$$\begin{aligned}F &= \frac{B^2 l^2 v}{r} & 0 \leq x < b \\ &= 0 & b \leq x < 2b\end{aligned}$$

The Joule heating loss is

$$\begin{aligned}P_J &= I^2 r \\ &= \frac{B^2 l^2 v^2}{r} & 0 \leq x < b \\ &= 0 & b \leq x < 2b\end{aligned}$$



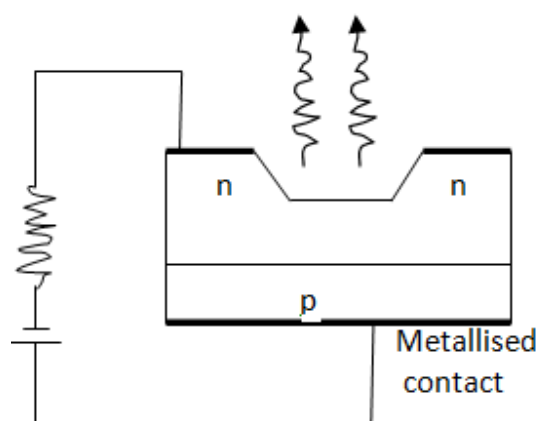
Q. 30. Part I: Being repeat question not answered in this answer sheet.

Part II: What is a light emitting diode (LED)? Mention two important advantages of LEDs over conventional lamps.

Answer:

b) Light Emitting Diode (LED)

These are forward biased p-n junction diode which emits spontaneous radiations. When electron falls from the higher to lower energy level containing holes, the energy is released in the form of light radiations.



It is a basic p-n-junction diode, which emits light when activated. When a fitting voltage is applied to the leads, electrons are able to recombine with electron holes within the device, that is when electron falls from higher to lower energy level containing holes the difference of energy is released in the form of photons (light). This effect is called electroluminescence, and the colour of the light (corresponding to the energy of the photon) is determined by the energy band gap of the semiconductor.

Advantages of LED: Two advantages over conventional incandescent lamps

- I. Low operational voltage and less power consumption.
- II. Fast action and no warm up time required.